

Midterm Exam 2

(11:05 AM – 12:20 PM : 75 Minutes)

- This exam will account for either 15% or 30% of your overall grade depending on your relative performance in midterm exam 1 and midterm exam 2. The higher of the two scores will be worth 30% of your grade, and the lower one 15%.
- There are four (4) questions worth 75 points in total. Please answer all of them in the spaces provided.
- There are twenty-two (22) pages, including nine (9) blank pages and two (2) pages of appendices. Please use the blank pages if you need additional space for your answers.
- The exam is **open slides** and **open notes** (including **scribe notes**). But **no books** and **no computers** are allowed.

GOOD LUCK!

Question	Pages	Parts	Points	Score
1. Akra-Bazzi and Generating Functions	2 – 6	(a) – (c)	$8 + 10 + 7 = 25$	
3. Conflict Reduction	8 – 10	(a) – (b)	$5 + 10 = 15$	
3. Randomized Quicksort	12 – 14	(a) – (b)	$5 + 10 = 15$	
3. A Simple Data Structure	16 – 18	(a) – (b)	$5 + 15 = 20$	
Total			75	

NAME: _____

SBU ID: _____

USELESS(n) $\{n = 2^m \text{ for some integer } m \geq 0\}$ 1. if $n = 1$ then $x \leftarrow 0$ 2. elseif $n = 2$ then $x \leftarrow 1$ 3. else 4. $u \leftarrow \text{USELESS}(\frac{n}{2})$ 5. $v \leftarrow \text{USELESS}(\frac{n}{4})$ 6. $w \leftarrow \text{USELESS}(\frac{n}{4})$ 7. $x \leftarrow u + v + w$ 8. return x

Figure 1: [Question 1] An inefficient function with unnecessary recursive calls which calculates something useless.

QUESTION 1. [25 Points] Akra-Bazzi and Generating Functions. The function USELESS given in Figure 1 computes and returns a value that is not very useful. It is known that its input n will always be a nonnegative power of 2, i.e., $n = 2^m$ for some integer $m \geq 0$, and that when called with such an input it returns a nonnegative integer not more than $m = \log_2 n$ bits long.

We assume that the machine registers are k bits long, and the machine can read from and write to these registers and add the contents of two such registers in $\mathcal{O}(1)$ time each. We also assume that USELESS will never be called with an n value that is too large to fit in a k -bit register.

This question asks you to compute the running time of USELESS(n).

- (a) [8 Points] Let $T(n)$ be the running time of USELESS(n). We claim that $T(n)$ can be described by the following recurrence relation, where we replace each $\Theta(1)$ term with a 1 for simplicity.

$$T(n) = \begin{cases} 1, & \text{if } n \leq 2, \\ T\left(\frac{n}{2}\right) + 2T\left(\frac{n}{4}\right) + 1, & \text{otherwise,} \end{cases}$$

Justify this recurrence and solve it using the Akra-Bazzi method.

Hint. Explain why every part of the recurrence makes sense based on the function given in Figure 1.

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- (b) [**10 Points**] Let $t_m = T(2^m) = T(n)$, where $T(n)$ is as defined in part (a). Clearly, t_m can be described by the following recurrence relation.

$$t_m = \begin{cases} 1, & \text{if } m \leq 1, \\ t_{m-1} + 2t_{m-2} + 1, & \text{otherwise,} \end{cases}$$

Let $F(z)$ be the ordinary generating function for t_m , i.e.,

$$F(z) = t_0 + t_1z + t_2z^2 + \dots + t_mz^m + \dots$$

Show that $F(z) = \frac{1}{2} \left(\frac{1}{1+z} - \frac{1}{1-z} \right) + \frac{1}{1-2z}$.

Hint. Try to solve it the way we solved the generating function for Fibonacci numbers in the class, not the way we solved the generating function for Quicksort. This question does not require you to do anything nearly as complicated as solving differential equations.

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(c) [**7 Points**] Use the result from part (b) to show that

$$t_m = 2^m - \frac{1}{2} \left(1 - (-1)^m \right)$$

Based on the solution for t_m , express $T(n)$ as a function of n .

Hint. $(1 - z)^{-1} = \sum_{m=0}^{\infty} z^m$.

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CREATE-GROUPS(  $f_1, f_2, \dots, f_n$  )
1.  $G_1 \leftarrow \emptyset, G_2 \leftarrow \emptyset$ 
2. for  $i \leftarrow 1$  to  $n$  do
3.    $s \leftarrow$  result of a fair coin flip
4.   if  $s = \text{HEADS}$  then  $G_1 \leftarrow G_1 \cup \{f_i\}$ 
5.   else  $G_2 \leftarrow G_2 \cup \{f_i\}$ 
6. return  $\langle G_1, G_2 \rangle$ 

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Figure 2: [Question 2] A randomized algorithm for dividing n people into two groups.

QUESTION 2. [15 Points] Conflict Reduction. You have n friends $f_1, f_2, \dots, f_n$. However, though they are all your friends, not everyone of them is in good terms with everyone else. If one of your friends f_i dislikes another friend f_j , you say that $\{f_i, f_j\}$ is a conflicting pair. Suppose that there are m such conflicting pairs.

Figure 2 shows a simple randomized algorithm called CREATE-GROUPS that divides your n friends into two groups G_1 and G_2 with an aim to reduce conflicts by splitting the conflicting pairs.

Let X be the number of conflicting pairs $\{f_i, f_j\}$ whose members are placed in different groups by CREATE-GROUPS.

(a) [5 Points] Calculate $E[X]$.

Hint. If $\{f_i, f_j\}$ is a conflicting pair, what is the probability that f_i and f_j end up in different groups?

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(b) [**10 Points**] Show that $Pr\left(X \leq \frac{m}{4}\right) \leq \frac{1}{e^{\frac{m}{16}}}$.

Hint. Use an appropriate Chernoff bound from the appendix.

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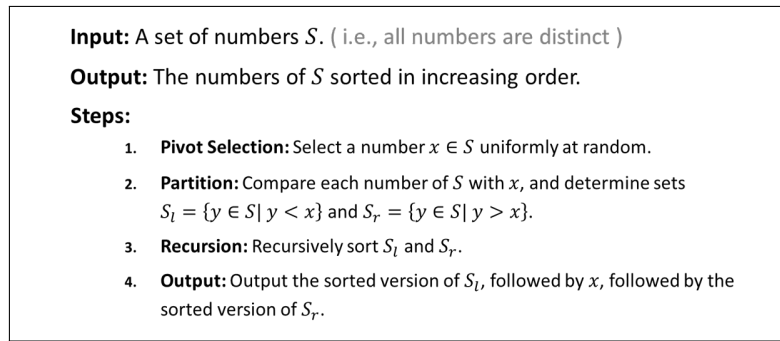


Figure 3: [Question 3] The randomized quicksort algorithm we saw in the class.

QUESTION 3. [15 Points] Randomized Quicksort. Figure 3 shows the randomized quicksort algorithm we saw in the class. This question asks you derive the expected running time of that algorithm.

- (a) [5 Points] Let $T(n)$ be the expected running time of the randomized quicksort given in Figure 3 on an input of size n . Explain why $T(n)$ can be described by the following recurrence.

$$T(n) \leq \begin{cases} \Theta(1), & \text{if } n \leq 8, \\ \frac{1}{2} \left(T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) \right) + \frac{1}{4} \left(T\left(\frac{n}{8}\right) + T\left(\frac{7n}{8}\right) \right) + \frac{1}{4}T(n) + \Theta(n), & \text{otherwise.} \end{cases}$$

Hint. If the randomized quicksort in Figure 3 chooses a pivot x that is neither among the smallest $\frac{n}{4}$ nor among the largest $\frac{n}{4}$ numbers in S , it will ensure that $\frac{n}{4} \leq |S_l|, |S_r| \leq \frac{3n}{4}$. What is the probability that such a pivot will be chosen? What if x is neither among the smallest $\frac{n}{8}$ nor among the largest $\frac{n}{8}$ numbers in S ?

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(b) [**10 Points**] Show that the recurrence given in part (a) can be rewritten as follows.

$$T(n) \leq \begin{cases} \Theta(1), & \text{if } n \leq 8, \\ \frac{2}{3} T\left(\frac{n}{4}\right) + \frac{2}{3} T\left(\frac{3n}{4}\right) + \frac{1}{3} T\left(\frac{n}{8}\right) + \frac{1}{3} T\left(\frac{7n}{8}\right) + \Theta(n), & \text{otherwise.} \end{cases}$$

Solve the recurrence.

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INIT()	<i>{initialize the data structure}</i>
1. $S \leftarrow \emptyset$	<i>{create an empty stack}</i>
2. PUSH(S, ∞)	<i>{push a sentinel item into the stack}</i>

INSERT(v)	<i>{insert the number v into the data structure}</i>
1. while TOP(S) $\leq v$ do	<i>{while the number at the top of the stack is $\leq v$}</i>
2. POP(S)	<i>{throw it away}</i>
3. PUSH(S, v)	<i>{push v into the stack}</i>

Figure 4: [Question 4] A simple data structure that supports only insertions.

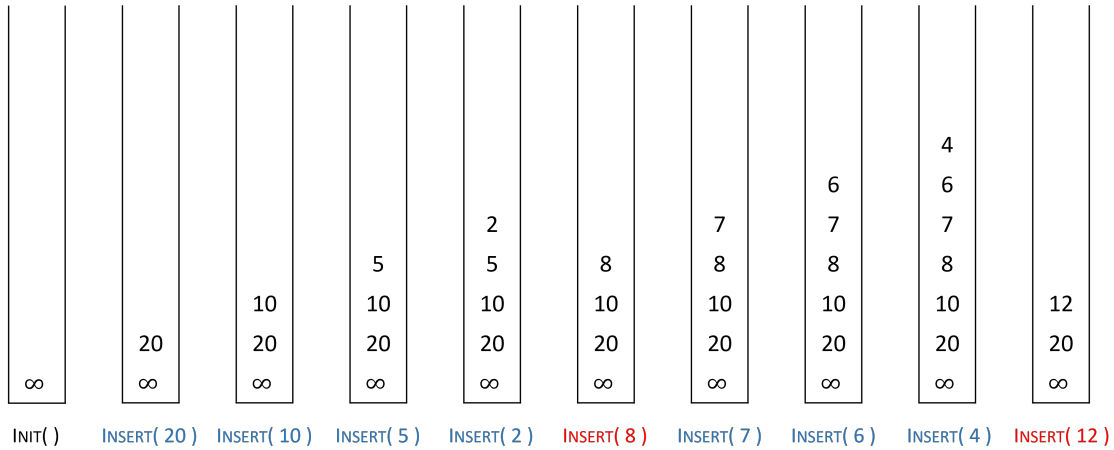


Figure 5: [Question 4] State of the data structure after each operation in a sequence of INSERTs.

QUESTION 4. [20 Points] A Simple Data Structure. Figure 4 shows the a simple data structure supports the following two operations.

- INIT() initializes the data structure, and
- INSERT(v) inserts the number v into the data structure.

Figure 5 shows how the state of the data structure changes after each operation in a sequence of INSERT operations.

- (a) [**5 Points**] What is the worst-case cost of an INSERT operation assuming n to be the number of items (including the sentinel) in the data structure right before the operation? Justify your answer.

- (b) [**15 Points**] Find the amortized cost of the INSERT operation using either the accounting method or the potential method.

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APPENDIX I: AKRA-BAZZI RECURRENCES

Consider the following recurrence:

$$T(x) = \begin{cases} \Theta(1), & \text{if } 1 \leq x \leq x_0, \\ \sum_{i=1}^k a_i T(b_i x) + g(x), & \text{otherwise,} \end{cases}$$

where,

1. $k \geq 1$ is an integer constant,
2. $a_i > 0$ is a constant for $1 \leq i \leq k$,
3. $b_i \in (0, 1)$ is a constant for $1 \leq i \leq k$,
4. $x \geq 1$ is a real number,
5. x_0 is a constant and $\geq \max \left\{ \frac{1}{b_i}, \frac{1}{1-b_i} \right\}$ for $1 \leq i \leq k$, and
6. $g(x)$ is a nonnegative function that satisfies a polynomial growth condition (e.g., $g(x) = x^\alpha \log^\beta x$ satisfies the polynomial growth condition for any constants $\alpha, \beta \in \mathbb{R}$).

Let p be the unique real number for which $\sum_{i=1}^k a_i b_i^p = 1$. Then $T(x) = \Theta \left(x^p \left(1 + \int_1^x \frac{g(u)}{u^{p+1}} du \right) \right)$.

APPENDIX II: USEFUL TAIL BOUNDS

Markov's Inequality. Let X be a random variable that assumes only nonnegative values. Then for all $\delta > 0$, $Pr[X \geq \delta] \leq \frac{E[X]}{\delta}$.

Chebyshev's Inequality. Let X be a random variable with a finite mean $E[X]$ and a finite variance $Var[X]$. Then for any $\delta > 0$, $Pr[|X - E[X]| \geq \delta] \leq \frac{Var[X]}{\delta^2}$.

Chernoff Bounds. Let $X_1, \dots, X_n$ be independent Poisson trials, that is, each X_i is a 0-1 random variable with $Pr[X_i = 1] = p_i$ for some p_i . Let $X = \sum_{i=1}^n X_i$ and $\mu = E[X]$. Following bounds hold:

Lower Tail:

- for $0 < \delta < 1$, $Pr[X \leq (1 - \delta)\mu] \leq \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}}\right)^\mu$
- for $0 < \delta < 1$, $Pr[X \leq (1 - \delta)\mu] \leq e^{-\frac{\mu\delta^2}{2}}$
- for $0 < \gamma < \mu$, $Pr[X \leq \mu - \gamma] \leq e^{-\frac{\gamma^2}{2\mu}}$

Upper Tail:

- for any $\delta > 0$, $Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}}\right)^\mu$
- for $0 < \delta < 1$, $Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$
- for $0 < \gamma < \mu$, $Pr[X \geq \mu + \gamma] \leq e^{-\frac{\gamma^2}{3\mu}}$